Profinite completions of products

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Abstract

A source of difficulty in profinite homotopy theory is that the profinite completion functor does not preserve finite products. In this note, we provide a new, checkable criterion on prospaces X and Y that guarantees that the profinite completion of $X \times Y$ agrees with the product of the profinite completions of X and Y. Using this criterion, we show that profinite completion preserves products of étale homotopy types of qcqs schemes. This fills a gap in Chough's proof of the Künneth formula for the étale homotopy type of a product of proper schemes over a separably closed field.

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0 Introduction

Write $\mathbf{Spc}_\pi$ for the ∞-category of π-finite spaces. Given a set Σ of primes, write $\mathbf{Spc}_\Sigma \subset \mathbf{Spc}_\pi$ for the full subcategory spanned by those π-finite spaces whose homotopy groups have orders divisible only by primes in Σ. Write

$$(-)^\wedge_\Sigma \colon \operatorname{Pro}(\mathbf{Spc}) \to \operatorname{Pro}(\mathbf{Spc}_\Sigma)$$

for the Σ-completion functor, i.e., the left adjoint to the inclusion $\operatorname{Pro}(\mathbf{Spc}_\Sigma) \subset \operatorname{Pro}(\mathbf{Spc})$. One source of difficulty in profinite homotopy theory is that the Σ-completion functor does not preserve finite limits, or even finite products (see [SAG, Remark E.5.2.6; 5, Remark 3.10]).

As far as we are aware, given connected spaces X and Y, the only general condition to check that the profinite completion of $X \times Y$ is the product of the profinite completions of X and Y is to check that the homotopy groups of X and Y are *good* in the sense of Serre [5, Proposition 3.9]. However, it is generally quite difficult to check if a group is good, and there are hard conjectures about whether or not certain groups are good. For example, Deligne and Morava's conjecture that the mapping class group $\Gamma_{g,n}$ of a genus g curve with n marked points is good [15, Problem on p. 94] is still open.

The purpose of this note is to provide a new, checkable criterion on prospaces X and Y that guarantees that the natural map $(X \times Y)^{\wedge}_{\Sigma} \to X^{\wedge}_{\Sigma} \times Y^{\wedge}_{\Sigma}$ is an equivalence. Since the statement of the this criterion requires introducing a bit of terminology, in this introduction we state the two applications that motivated our general results. For the precise statements of the general results, see Corollary 2.8 and Theorem 2.13.

The first application is that if one is already in the setting of profinite homotopy theory, then Σ-completion preserves products:

0.1 Proposition (Corollary 2.9)**.** *Let Σ be a set of primes. Then the Σ-completion functor restricted to profinite spaces*

$$(-)^{\wedge}_{\Sigma} \colon \operatorname{Pro}(\mathbf{Spc}_{\pi}) \to \operatorname{Pro}(\mathbf{Spc}_{\Sigma})$$

preserves products.

The second is that in the setting of étale homotopy theory, Σ-completion preserves finite products. Given a scheme X, write $\Pi^{\text{ét}}_{\infty}(X) \in \operatorname{Pro}(\mathbf{Spc})$ for the étale homotopy type of X.

0.2 Proposition (Example 2.17)**.** *Let Σ be a set of primes and let X and Y be qcqs schemes. Then the natural map of profinite spaces*

$$\big(\Pi^{\text{ét}}_{\infty}(X) \times \Pi^{\text{ét}}_{\infty}(Y)\big)^{\wedge}_{\Sigma} \to \Pi^{\text{ét}}_{\infty}(X)^{\wedge}_{\Sigma} \times \Pi^{\text{ét}}_{\infty}(Y)^{\wedge}_{\Sigma}$$

is an equivalence.

0.3 Remark. Proposition 0.2 fills a gap in Chough's proof of the Künneth formula for the étale homotopy type of a product of proper schemes over a separably closed field [7, Theorem 5.3]. Chough's proof cites the false claim that profinite completion preserves finite limits. However, what Chough actually uses is Proposition 0.2 (with Σ the set of all primes). In particular, the conclusion of [7, Theorem 5.3] remains valid.

We also remark that in our work with Holzschuh and Wolf [8, §4], Proposition 0.2 is a key ingredient used to prove other Künneth formulas in étale homotopy theory.

0.4 Proof Strategy. Propositions 0.1 and 0.2 are consequences of a more general result. To explain why this is the case, first note that since Σ-completion preserves cofiltered limits, to prove Proposition 0.1 it suffices to show that Σ-completion preserves finite products of π-finite spaces. This reduction is useful because π-finite spaces admit very nice presentations: every π-finite space can be written as the geometric realization (in **Spc**) of a Kan complex with finitely many simplices in each dimension [SAG, Lemma E.1.6.5].

Similarly, to prove Proposition 0.2 we use that the étale homotopy type of a qcqs scheme admits a nice presentation. To see this, the first technical observation is that since protruncation preserves limits [9, Proposition 3.9] and profinite completion factors through protruncation, it suffices to replace the étale homotopy types by their protruncations. Our work with Barwick and Glasman [1, Theorems 10.2.3 & 12.5.1] provides a description of the protruncated étale homotopy type as the protruncated classifying space of an explicit profinite category. Said differently, the protruncated étale homotopy type can be written as a geometric realization of a simplicial profinite space computed in the larger ∞-category of protruncated spaces (see Example A.9).

Hence we're done if we can prove the more general claim that Σ-completion preserves products of protruncated spaces that admit such presentations; this is our main result, see Theorem 2.13. This follows once we know that that geometric realizations preserve finite products in the ∞-categories of protruncated and Σ-profinite spaces (see Proposition 1.17 and Corollary 1.18). See Lemma 2.7 and Corollary 2.8 for the key categorical results that we use.

0.5 Linear Overview. Section 1 proves that geometric realizations are universal in the ∞-categories of protruncated and profinite spaces. In particular, geometric realizations preserve finite products in these ∞-categories. Section 2 proves Propositions 0.1 and 0.2. It is immediate from [1, Theorem 10.2.3] that the protruncated étale homotopy type can be written as the geometric realization of a simplicial profinite space. However, for ease of reference we have provided a detailed explanation of this fact in Appendix A.

0.6 Conventions. Throughout, we use the notational conventions of [9, §§1 & 3]. In an effort to keep this note short, we do not recapitulate them here.

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1 Universality of colimits

In this section, we prove that geometric realizations are universal in the ∞-categories of protruncated and Σ-profinite spaces. We accomplish this by proving a more general fact: colimits over diagrams that can be computed as finite colimits when valued in an n-category (see Definition 1.9) are universal in the ∞-categories of protruncated and Σ-profinite spaces (Proposition 1.17 and Corollary 1.18).

The first observation is that finite colimits are universal in protruncated spaces.

1.1 Lemma. *Let $\mathcal{C}$ be an ∞-category with pullbacks and finite colimits. If finite colimits are universal in $\mathcal{C}$, then finite colimits are universal in* $\mathrm{Pro}(\mathcal{C})$*.*

Proof. By (the dual of) [HTT, Proposition 5.3.5.15], pullbacks, pushouts, and finite coproducts are computed 'levelwise' in $\mathrm{Pro}(\mathcal{C})$. Thus the assumption that finite colimits are universal in $\mathcal{C}$ implies the claim. □

1.2 Example. Finite colimits are universal in $\mathrm{Pro}(\mathbf{Spc})$. For each integer $n \geq 0$, finite colimits are universal in $\mathrm{Pro}(\mathbf{Spc}_{\leq n})$.

1.3 Recollection. A localization $L\colon \mathcal{C} \to \mathcal{D}$ is *locally cartesian* if for any cospan $X \to Z \leftarrow Y$ such that $X, Z \in \mathcal{D}$, the natural map $L(X \times_Z Y) \to X \times_Z L(Y)$ is an equivalence.

1.4 Example [9, Proposition 3.18]**.** For any set Σ of primes, the localization

$$(-)^{\wedge}_{\Sigma}\colon\ \mathrm{Pro}(\mathbf{Spc}_{<\infty}) \to \mathrm{Pro}(\mathbf{Spc}_{\Sigma})$$

is locally cartesian. However, $(-)^{\wedge}_{\Sigma}$ does not generally preserve finite products.

The following is immediate from the definitions:

1.5 Lemma. *Let $\mathcal{I}$ be an ∞-category, $\mathcal{C}$ an ∞-category with pullbacks and $\mathcal{I}$-shaped colimits, and let $L\colon \mathcal{C} \to \mathcal{D}$ be a locally cartesian localization. If $\mathcal{I}$-shaped colimits are universal in $\mathcal{C}$, then $\mathcal{I}$-shaped colimits are universal in $\mathcal{D}$.*

1.6 Example. Since the protruncation functor $\tau_{<\infty} : \mathrm{Pro}(\mathbf{Spc}) \to \mathrm{Pro}(\mathbf{Spc}_{<\infty})$ preserves limits [9, Proposition 3.9], Example 1.2 and Lemma 1.5 show that finite colimits are universal in $\mathrm{Pro}(\mathbf{Spc}_{<\infty})$.

Now we formulate the key property of the category $\mathbf{\Delta}^{\mathrm{op}}$ that we need.

1.7 Definition. Let $n \geq 0$ be an integer. A functor between ∞-categories $c : \mathcal{I} \to \mathcal{J}$ is *n-colimit-cofinal* if for every n-category $\mathcal{C}$ and functor $f : \mathcal{J} \to \mathcal{C}$, the following conditions are satisfied:

(1.7.1) The colimit $\mathrm{colim}_{\mathcal{J}} f$ exists if and only if the colimit $\mathrm{colim}_{\mathcal{I}} fc$ exists.

(1.7.2) If the colimit $\mathrm{colim}_{\mathcal{J}} f$ exists, then the natural map $\mathrm{colim}_{\mathcal{I}} fc \to \mathrm{colim}_{\mathcal{J}} f$ is an equivalence.

1.8 Example. For an integer $n \geq 0$, write $\mathbf{\Delta}_{\leq n} \subset \mathbf{\Delta}$ for the full subcategory spanned by those nonempty linearly ordered finite sets of cardinality $\leq n+1$. By [10, Proposition A.1], the inclusion $\mathbf{\Delta}_{\leq n}^{\mathrm{op}} \subset \mathbf{\Delta}^{\mathrm{op}}$ is n-colimit-cofinal.

1.9 Definition. Let $\mathcal{I}$ be an ∞-category. We say that $\mathcal{I}$ is *almost finite* if for each integer $n \geq 0$, there exists a *finite* ∞-category $\mathcal{I}_n$ and an n-colimit-cofinal functor $c_n : \mathcal{I}_n \to \mathcal{I}$.

Here are a number of important examples of almost finite ∞-categories.

1.10 Example. If $\mathcal{I}$ is an ∞-category that admits a colimit-cofinal functor from a finite ∞-category, then $\mathcal{I}$ is almost finite.

1.11 Example. For each $n \geq 0$, the category $\mathbf{\Delta}_{\leq n}^{\mathrm{op}}$ is a finite ∞-category [6, Example 6.5.3]. Hence the category $\mathbf{\Delta}^{\mathrm{op}}$ is almost finite: the inclusion $\mathbf{\Delta}_{\leq n}^{\mathrm{op}} \hookrightarrow \mathbf{\Delta}^{\mathrm{op}}$ is an n-colimit-cofinal functor from a finite ∞-category.

1.12 Definition. Let K be a simplicial set. The *∞-category presented by K* is the image of K under the natural functor $\mathrm{s}\mathbf{Set} \to \mathbf{Cat}_{\infty}$ obtained by inverting the weak equivalences in the Joyal model structure. The *space presented by K* is the image of K under the natural functor $\mathrm{s}\mathbf{Set} \to \mathbf{Spc}$ obtained by inverting the weak equivalences in the Kan–Quillen model structure.

1.13 Example. Let K be a simplicial set with finitely many simplices in each dimension and let $\mathcal{I}$ be the ∞-category presented by K. Then $\mathcal{I}$ is almost finite: we take $\mathcal{I}_n$ to be the ∞-category presented by the $(n+1)$-skeleton of $\mathrm{sk}_{n+1} K$ and $c_n : \mathcal{I}_n \to \mathcal{I}$ the functor induced by the inclusion $\mathrm{sk}_{n+1} K \subset K$.

1.14 Recollection. A space X is *almost π-finite* if $\pi_0(X)$ is finite and all homotopy groups of X are finite. An almost π-finite space admits a presentation by a Kan complex with finitely many simplices in each dimension [SAG, Lemma E.1.6.5]. Hence:

1.15 Example. As a special case of Example 1.13, every amost π-finite space is an almost finite ∞-category.

1.16 Definition. Let $\mathcal{C}$ be an ∞-category with pullbacks. We say that *almost finite colimits are universal in $\mathcal{C}$* if for each almost finite ∞-category $\mathcal{I}$, the ∞-category $\mathcal{C}$ admits $\mathcal{I}$-shaped colimits and $\mathcal{I}$-shaped colimits are universal in $\mathcal{C}$.

The argument that Lurie gives in the proof of [SAG, Theorem E.6.3.1] essentially shows that almost finite colimits are universal in $\mathrm{Pro}(\mathbf{Spc}_{\Sigma})$. However, the statement about universality of colimits is less general and the result is only stated when Σ is the set of all primes. We also need to know that almost finite colimits are also universal in $\mathrm{Pro}(\mathbf{Spc}_{<\infty})$. The strategy is the same as Lurie's proof: we use that equivalences are checked on truncations to reduce to the case of finite colimits.

1.17 Proposition. *Almost finite colimits are universal in* $\mathrm{Pro}(\mathbf{Spc}_{<\infty})$.

Moreover, Proposition 1.17 reproves and strengthens [SAG, Theorem E.6.3.1]:

1.18 Corollary. *Let* Σ *be a set of primes. Then almost finite colimits are universal in* $\mathrm{Pro}(\mathbf{Spc}_{\Sigma})$.

Proof of Corollary 1.18. Since the localization $(-)^{\wedge}_{\Sigma} \colon \mathrm{Pro}(\mathbf{Spc}_{<\infty}) \to \mathrm{Pro}(\mathbf{Spc}_{\Sigma})$ is locally cartesian [9, Proposition 3.18], this follows from Lemma 1.5 and Proposition 1.17. ☐

Proof of Proposition 1.17. Let $\mathcal{I}$ be an almost finite ∞-category, let $f \colon X \to Z$ be a morphism in $\mathrm{Pro}(\mathbf{Spc}_{<\infty})$, and let

$$g \colon \mathcal{I} \to \mathrm{Pro}(\mathbf{Spc}_{<\infty})_{/Z}$$

be a diagram of protruncated spaces over Z. To prove the claim, it suffices to show that for each integer $n \geq 0$, the induced map

$$\tau_{\leq n}\left(\operatorname*{colim}_{i \in \mathcal{I}} X \times_Z g(i)\right) \to \tau_{\leq n}\left(X \times_Z \operatorname*{colim}_{i \in \mathcal{I}} g(i)\right)$$

is an equivalence in $\mathrm{Pro}(\mathbf{Spc}_{\leq n})$. Since $\mathcal{I}$ is almost finite, there exists a finite ∞-category $\mathcal{I}_{n+2}$ and $(n+2)$-colimit-cofinal functor $c_{n+2} \colon \mathcal{I}_{n+2} \to \mathcal{I}$. Consider the commutative diagram

$$\begin{tikzcd}
\operatorname*{colim}_{j \in \mathcal{I}_{n+2}} \tau_{\leq n}\left(X \times_Z gc_{n+2}(j)\right) \arrow[r, "\sim"] \arrow[d] & \tau_{\leq n}\left(\operatorname*{colim}_{j \in \mathcal{I}_{n+2}} X \times_Z gc_{n+2}(j)\right) \arrow[r] \arrow[d] & \tau_{\leq n}\left(X \times_Z \operatorname*{colim}_{j \in \mathcal{I}_{n+2}} gc_{n+2}(j)\right) \arrow[d] \\
\operatorname*{colim}_{i \in \mathcal{I}} \tau_{\leq n}\left(X \times_Z g(i)\right) \arrow[r, "\sim"] & \tau_{\leq n}\left(\operatorname*{colim}_{i \in \mathcal{I}} X \times_Z g(i)\right) \arrow[r] & \tau_{\leq n}\left(X \times_Z \operatorname*{colim}_{i \in \mathcal{I}} g(i)\right) .
\end{tikzcd}$$

(Here, the colimits in the leftmost column are computed in $\mathrm{Pro}(\mathbf{Spc}_{\leq n})$.) Since $\mathrm{Pro}(\mathbf{Spc}_{\leq n})$ is an $(n+1)$-category and $c_{n+2} \colon \mathcal{I}_{n+2} \to \mathcal{I}$ is $(n+2)$-colimit-cofinal, the leftmost vertical map is an equivalence. Thus the central vertical map is also an equivalence. Since $\mathcal{I}_{n+2}$ is finite and finite colimits are universal in $\mathrm{Pro}(\mathbf{Spc}_{<\infty})$ (Example 1.6), the top right-hand horizontal map is an equivalence.

To complete the proof, it suffices to show that the rightmost vertical map is an equivalence. For this, consider the commutative square

$$\begin{tikzcd}
\operatorname*{colim}_{j \in \mathcal{I}_{n+2}} \tau_{\leq n+1}\, gc_{n+2}(j) \arrow[r, "\sim"] \arrow[d] & \tau_{\leq n+1}\left(\operatorname*{colim}_{j \in \mathcal{I}_{n+2}} gc_{n+2}(j)\right) \arrow[d] \\
\operatorname*{colim}_{i \in \mathcal{I}} \tau_{\leq n+1}\, g(i) \arrow[r, "\sim"] & \tau_{\leq n+1}\left(\operatorname*{colim}_{i \in \mathcal{I}} g(i)\right) ,
\end{tikzcd}$$

where the colimits in the left-hand column are computed in $\mathrm{Pro}(\mathbf{Spc}_{\leq n+1})$. Since $\mathrm{Pro}(\mathbf{Spc}_{\leq n+1})$ is an $(n+2)$-category and c_{n+2} is $(n+2)$-colimit-cofinal, the left-hand vertical map is an equivalence. Hence the right-hand vertical map is also an equivalence. As a consequence, the map

$$\operatorname*{colim}_{j \in \mathcal{I}_{n+2}} gc_{n+2}(j) \to \operatorname*{colim}_{i \in \mathcal{I}} g(i)$$

is n-connected. Thus the basechange

$$X \underset{Z}{\times} \operatorname*{colim}_{j \in \mathcal{I}_{n+2}} gc_{n+2}(j) \to X \underset{Z}{\times} \operatorname*{colim}_{i \in \mathcal{I}} g(i)$$

is also n-connected. Hence the map

$$\tau_{\leq n}\left(X \underset{Z}{\times} \operatorname*{colim}_{j \in \mathcal{I}_{n+2}} gc_{n+2}(j)\right) \to \tau_{\leq n}\left(X \underset{Z}{\times} \operatorname*{colim}_{i \in \mathcal{I}} g(i)\right)$$

is an equivalence, as desired. □

Note that the universality of geometric realizations implies that geometric realizations preserve finite products:

1.19 Lemma. *Let $\mathcal{J}$ be a sifted ∞-category and let $\mathcal{C}$ be an ∞-category with finite limits and $\mathcal{J}$-shaped colimits. If $\mathcal{J}$-shaped colimits are universal in $\mathcal{C}$, then the functor*

$$\operatorname{colim}_{\mathcal{J}} \colon \operatorname{Fun}(\mathcal{J}, \mathcal{C}) \to \mathcal{C}$$

preserves finite products.

Proof. Let $X_\bullet, Y_\bullet \colon \mathcal{J} \to \mathcal{C}$ be functors. We have natural equivalences

$$\begin{aligned}
\operatorname*{colim}_{i \in \mathcal{J}} X_i \times Y_i &\xrightarrow{\sim} \operatorname*{colim}_{(i,j) \in \mathcal{J} \times \mathcal{J}} X_i \times Y_j && (\mathcal{J} \text{ is sifted}) \\
&\simeq \operatorname*{colim}_{i \in \mathcal{J}} \operatorname*{colim}_{j \in \mathcal{J}} (X_i \times Y_j) \\
&\xrightarrow{\sim} \operatorname*{colim}_{i \in \mathcal{J}} \left(X_i \times \operatorname*{colim}_{j \in \mathcal{J}} Y_j\right) && (\mathcal{J}\text{-shaped colimits are universal}) \\
&\xrightarrow{\sim} \left(\operatorname*{colim}_{i \in \mathcal{J}} X_i\right) \times \left(\operatorname*{colim}_{j \in \mathcal{J}} Y_j\right) && (\mathcal{J}\text{-shaped colimits are universal})\,.
\end{aligned}$$

□

1.20 Corollary. *Let Σ be a set of primes. Then geometric realizations preserve finite products in the ∞-categories* $\operatorname{Pro}(\mathbf{Spc}_\Sigma)$ *and* $\operatorname{Pro}(\mathbf{Spc}_{<\infty})$.

Proof. Combine Proposition 1.17, Corollary 1.18, and Lemma 1.19. □

2 Completions of products

The goal of this section is to prove Propositions 0.1 and 0.2. We do this by noting that in the setting of both of these results, the prospaces of interest can be written as geometric realizations of simplicial profinite spaces. We begin by axiomatizing the situation:

2.1 Definition. Let $\mathcal{C}$ be an ∞-category, let $\mathcal{D} \subset \mathcal{C}$ be a full subcategory, and let $X \in \mathcal{C}$. A *$\mathcal{D}$-resolution* of X is a simplicial object $X_\bullet \colon \mathbf{\Delta}^{\mathrm{op}} \to \mathcal{D}$ together with an equivalence

$$X \simeq \operatorname{colim}\left(\mathbf{\Delta}^{\mathrm{op}} \xrightarrow{X_\bullet} \mathcal{D} \hookrightarrow \mathcal{C}\right).$$

We say that an object X *admits a $\mathcal{D}$-resolution* if there exists $\mathcal{D}$-resolution of X.

2.2 Notation. Write $\mathbf{Set}_{\mathrm{fin}}$ for the category of a finite sets and all maps.

Finite and almost π-finite spaces admit $\mathbf{Set}_{\mathrm{fin}}$-resolutions:

2.3 Lemma. *Let X be a space which is finite*[1] *or almost π-finite. Then:*

(1) *As an object of* $\mathbf{Spc}$*, the space X admits a* $\mathbf{Set}_{\mathrm{fin}}$*-resolution.*

(2) *When regarded as a constant object of* $\mathrm{Pro}(\mathbf{Spc})$*, the space X admits a* $\mathbf{Set}_{\mathrm{fin}}$*-resolution.*

(3) *The protruncated space $\tau_{<\infty}(X)$ admits a* $\mathbf{Set}_{\mathrm{fin}}$*-resolution.*

(4) *The profinite space $X_{\pi}^{\wedge}$ admits a* $\mathbf{Set}_{\mathrm{fin}}$*-resolution.*

Proof. For (1), if X is finite, then X can be written as the geometric realization of a simplicial set with finitely many nondegenerate simplices [17, Proposition 2.4].[2] If X is almost π-finite, then X can be written as the geometric realization of a Kan complex with finitely many simplices in each dimension [SAG, Lemma E.1.6.5].

Now note that (2)–(4) follow from (1) and that each functor in the diagram

$$\mathbf{Spc} \hookrightarrow \mathrm{Pro}(\mathbf{Spc}) \xrightarrow{\tau_{<\infty}} \mathrm{Pro}(\mathbf{Spc}_{<\infty}) \xrightarrow{(-)_{\pi}^{\wedge}} \mathrm{Pro}(\mathbf{Spc}_{\pi})$$

preserves colimits. □

Algebraic geometry also gives rise to many examples of protruncated spaces admitting profinite resolutions.

2.4 Notation (shapes)**.** Given an ∞-topos $\boldsymbol{X}$, we write $\Pi_{\infty}(\boldsymbol{X}) \in \mathrm{Pro}(\mathbf{Spc})$ for the *shape* of $\boldsymbol{X}$. We write $\Pi_{<\infty}(\boldsymbol{X})$ for the protruncation of $\Pi_{\infty}(\boldsymbol{X})$.

2.5. Recall that an ∞-topos $\boldsymbol{X}$ is *spectral* in the sense of [1, Definition 9.2.1] if $\boldsymbol{X}$ is bounded coherent and the ∞-category $\mathrm{Pt}(\boldsymbol{X})$ of points of $\boldsymbol{X}$ has the property that every endomorphism of an object of $\mathrm{Pt}(\boldsymbol{X})$ is an equivalence. The most important example of a spectral ∞-topos is the étale ∞-topos of a qcqs scheme [1, Example 9.2.4]. Example A.9 explains why the protruncated shape $\Pi_{<\infty}(\boldsymbol{X})$ of a spectral ∞-topos admits a natural $\mathrm{Pro}(\mathbf{Spc}_{\pi})$-resolution.

The next few results are the key categorical input we need. To state them, we axiomatize the abstract properties of the subcategory $\mathrm{Pro}(\mathbf{Spc}_{\pi}) \subset \mathrm{Pro}(\mathbf{Spc}_{<\infty})$.

2.6 Notation. Let $\mathcal{C}$ be an ∞-category with geometric realizations and finite products, and let $L \colon \mathcal{C} \to \mathcal{D}$ be a localization. Assume that geometric realizations preserve finite products in both $\mathcal{C}$ and $\mathcal{D}$. Write

$$\mathcal{C}_{|\mathcal{D}|} \subset \mathcal{C}$$

for the smallest full subcategory containing $\mathcal{D} \subset \mathcal{C}$ and closed under geometric realizations and retracts.

[1] I.e., in the smallest subcategory of $\mathbf{Spc}$ containing the point and the empty set and closed under pushouts. Equivalently, a space X is finite if and only if X is represented by a finite CW complex.

[2] In fact, every finite space is equivalent to the classifying space of a finite poset; see [HTT, Proposition 4.1.1.3(3) & Variant 4.2.3.16; Ker, Tag 02MU; 14, Theorem 1].

2.7 Lemma. *In the setting of Notation 2.6, let $X_\bullet, Y_\bullet \colon \mathbf{\Delta}^{\mathrm{op}} \to \mathcal{C}$ be simplicial objects with colimits X and Y. Assume that for each $n \geq 0$, the natural map*

$$L(X_n \times Y_n) \to L(X_n) \times L(Y_n)$$

is an equivalence. Then the natural map

$$L(X \times Y) \to L(X) \times L(Y)$$

is an equivalence.

Proof. We compute

$$\begin{aligned}
L(X \times Y) &\simeq L\left(\operatorname*{colim}_{m \in \mathbf{\Delta}^{\mathrm{op}}} X_m \times \operatorname*{colim}_{n \in \mathbf{\Delta}^{\mathrm{op}}} Y_n\right) & \\
&\simeq L\left(\operatorname*{colim}_{n \in \mathbf{\Delta}^{\mathrm{op}}} X_n \times Y_n\right) & \text{(geometric realizations preserve finite products in } \mathcal{C}) \\
&\simeq \operatorname*{colim}_{n \in \mathbf{\Delta}^{\mathrm{op}}} L(X_n \times Y_n) & \\
&\xrightarrow{\sim} \operatorname*{colim}_{n \in \mathbf{\Delta}^{\mathrm{op}}} L(X_n) \times L(Y_n) & \text{(assumption)} \\
&\simeq \operatorname*{colim}_{m \in \mathbf{\Delta}^{\mathrm{op}}} L(X_m) \times \operatorname*{colim}_{n \in \mathbf{\Delta}^{\mathrm{op}}} L(Y_n) & \text{(geometric realizations preserve finite products in } \mathcal{D}) \\
&\simeq L(X) \times L(Y)\,. &
\end{aligned}$$

□

2.8 Corollary. *In the setting of Notation 2.6:*

(1) *The full subcategory of $\mathcal{C} \times \mathcal{C}$ spanned by those objects (X, Y) such that the natural map*

$$L(X \times Y) \to L(X) \times L(Y)$$

is an equivalence is closed under geometric realizations and retracts.

(2) *If $X, Y \in \mathcal{C}_{|\mathcal{D}|}$, then the natural map $L(X \times Y) \to L(X) \times L(Y)$ is an equivalence.*

(3) *If $X, Y \in \mathcal{C}$ admit $\mathcal{D}$-resolutions, then the natural map $L(X \times Y) \to L(X) \times L(Y)$ is an equivalence.*

Proof. Item (1) follows from Lemma 2.7 and the fact that equivalences are closed under retracts. Now (2) is an immediate consequence of (1) and the definition of $\mathcal{C}_{|\mathcal{D}|}$ as the closure of $\mathcal{D} \subset \mathcal{C}$ under geometric realizations and retracts. Finally, (3) is a special case of (2). □

We now record some consequences of Corollary 2.8.

2.9 Corollary. *Let Σ be a set of prime numbers. Then the Σ-completion functor*

$$(-)^{\wedge}_{\Sigma} \colon \operatorname{Pro}(\mathbf{Spc}_{\pi}) \to \operatorname{Pro}(\mathbf{Spc}_{\Sigma})$$

preserves products.

Proof. Since Σ-completion preserves cofiltered limits and the terminal object, it suffices to show that Σ-completion preserves binary products of profinite spaces. Again because Σ-completion preserves cofiltered limits, we are reduced to showing that if X and Y are π-finite spaces, then the natural map

$$(X \times Y)^{\wedge}_{\Sigma} \to X^{\wedge}_{\Sigma} \times Y^{\wedge}_{\Sigma}$$

is an equivalence. Since π-finite spaces admit $\mathbf{Set}_{\mathrm{fin}}$-resolutions (Lemma 2.3) and geometric realizations preserve finite products in $\mathrm{Pro}(\mathbf{Spc}_{\pi})$ and $\mathrm{Pro}(\mathbf{Spc}_{\Sigma})$ (Corollary 1.20), the claim follows from Corollary 2.8. ☐

2.10 Warning. The functor $(-)^{\wedge}_{\Sigma} \colon \mathrm{Pro}(\mathbf{Spc}_{\pi}) \to \mathrm{Pro}(\mathbf{Spc}_{\Sigma})$ does not generally preserve pullbacks, or even loop objects.

We now introduce a slight enlargement of the subcategory of protruncated spaces admitting a $\mathrm{Pro}(\mathbf{Spc}_{\pi})$-resolution on which profinite completion preserves finite products.

2.11 Notation. Let

$$\mathrm{Pro}(\mathbf{Spc}_{<\infty})' \subset \mathrm{Pro}(\mathbf{Spc}_{<\infty})$$

denote the smallest full subcategory containing $\mathrm{Pro}(\mathbf{Spc}_{\pi})$ and closed under geometric realizations, retracts, and cofiltered limits.

2.12 Observation (procompact spaces)**.** Write $\mathbf{Spc}^{\omega} \subset \mathbf{Spc}$ for the full subcategory spanned by the compact objects.[3] Then by Lemma 2.3, the image of

$$\tau_{<\infty} \colon \mathrm{Pro}(\mathbf{Spc}^{\omega}) \to \mathrm{Pro}(\mathbf{Spc}_{<\infty})$$

is contained in $\mathrm{Pro}(\mathbf{Spc}_{<\infty})'$.

The following is the main result of this note.

2.13 Theorem. *Let Σ be a set of primes and let $X, Y \in \mathrm{Pro}(\mathbf{Spc}_{<\infty})'$. Then the natural map*

$$(X \times Y)^{\wedge}_{\Sigma} \to X^{\wedge}_{\Sigma} \times Y^{\wedge}_{\Sigma}$$

is an equivalence.

Proof. By Corollary 2.9, the Σ-completion functor

$$(-)^{\wedge}_{\Sigma} \colon \mathrm{Pro}(\mathbf{Spc}_{\pi}) \to \mathrm{Pro}(\mathbf{Spc}_{\Sigma})$$

preserves products. Hence it suffices to prove the claim in the special case where Σ is the set of all primes. For this, note that cofiltered limits preserve finite products in $\mathrm{Pro}(\mathbf{Spc}_{<\infty})$ and $\mathrm{Pro}(\mathbf{Spc}_{\pi})$; moreover, the profinite completion functor

$$(-)^{\wedge}_{\pi} \colon \mathrm{Pro}(\mathbf{Spc}_{<\infty}) \to \mathrm{Pro}(\mathbf{Spc}_{\pi})$$

preserves cofiltered limits. Thus it suffices to treat the case where X and Y are in the smallest full subcategory of $\mathrm{Pro}(\mathbf{Spc}_{<\infty})$ containing $\mathrm{Pro}(\mathbf{Spc}_{\pi})$ and closed under geometric realizations and retracts. In this case, since geometric realizations preserve finite products in $\mathrm{Pro}(\mathbf{Spc}_{<\infty})$ and $\mathrm{Pro}(\mathbf{Spc}_{\pi})$ (Corollary 1.20), Corollary 2.8 completes the proof. ☐

2.14 Example. If X and Y are protruncated spaces that admit $\mathrm{Pro}(\mathbf{Spc}_{\pi})$-resolutions, then the natural map $(X \times Y)^{\wedge}_{\Sigma} \to X^{\wedge}_{\Sigma} \times Y^{\wedge}_{\Sigma}$ is an equivalence.

[3] A space X is compact if and only if X is a retract of a finite space. In more classical terminology, a space X is compact if and only if X is represented by a *fintiely dominated* CW complex.

2.15 Example. If $X, Y \in \mathbf{Spc}$ are compact, then the natural map $(X \times Y)^\wedge_\Sigma \to X^\wedge_\Sigma \times Y^\wedge_\Sigma$ is an equivalence.

We conclude by recording two applications of Theorem 2.13.

2.16 Corollary. *Let Σ be a set of primes and let $\boldsymbol{X}$ and $\boldsymbol{Y}$ be spectral ∞-topoi. Then the natural map*

$$(\Pi_\infty(\boldsymbol{X}) \times \Pi_\infty(\boldsymbol{Y}))^\wedge_\Sigma \to \Pi_\infty(\boldsymbol{X})^\wedge_\Sigma \times \Pi_\infty(\boldsymbol{Y})^\wedge_\Sigma$$

is an equivalence.

Proof. Example A.9 shows that $\Pi_{<\infty}(\boldsymbol{X})$ and $\Pi_{<\infty}(\boldsymbol{Y})$ admit $\mathrm{Pro}(\mathbf{Spc}_\pi)$-resolutions. Since protruncation preserves products, the claim now follows from Theorem 2.13. □

2.17 Example. Let Σ be a set of primes and let X and Y be qcqs schemes. Then the natural map

$$\left(\Pi_\infty^{\text{ét}}(X) \times \Pi_\infty^{\text{ét}}(Y)\right)^\wedge_\Sigma \to \Pi_\infty^{\text{ét}}(X)^\wedge_\Sigma \times \Pi_\infty^{\text{ét}}(Y)^\wedge_\Sigma$$

is an equivalence.

The second application is to ∞-operads. Recently, it has become clear that it is important to consider profinite completions of ∞-operads. For example, Boavida de Brito, Horel, and Robertson explained a beautiful relationship between the Grothendieck–Teichmüller group and the profinite completion of the genus 0 surface operad [5]. However, since profinite completion does not preserve products, profinite completions of ∞-operads generally cannot be computed 'levelwise'. In general, more care is required to work with profinite completions of ∞-operads; this is the topic of recent work of Blom and Moerdijk [3; 4].

An immediate consequence of Theorem 2.13 is that profinite completions *can* be computed levelwise more generally than was previously known:

2.18 Corollary. *Let Σ be a set of primes and let $\{\mathcal{O}(n)\}_{n\geq 0}$ be an ∞-operad in spaces. Assume that for each $n \geq 0$, we have*

$$\tau_{<\infty}\, \mathcal{O}(n) \in \mathrm{Pro}(\mathbf{Spc}_{<\infty})' \,.$$

Then the levelwise Σ-completion $\{\mathcal{O}(n)^\wedge_\Sigma\}_{n\geq 0}$ defines an ∞-operad in $\mathrm{Pro}(\mathbf{Spc}_\Sigma)$.

A Classifying prospaces via geometric realizations

The purpose of this appendix is to explain why the protruncated shape of a spectral ∞-topos (e.g., the étale ∞-topos of a qcqs scheme) admits a presentation as a geometric realization of a simplicial profinite space (Example A.9). Using the description of the protruncated shape of a spectral ∞-topos as a protruncated classifying prospace given by Barwick–Glasman–Haine [1, Theorem 10.2.3], this is an exercise in the definitions. For the ease of the reader, we spell out the details here.

We make use of the description of ∞-categories as simplicial spaces.

A.1 Recollection (∞-categories as simplicial spaces)**.** The nerve functor

$$\begin{aligned} \mathrm{N}\colon \mathbf{Cat}_\infty &\to \mathrm{Fun}(\boldsymbol{\Delta}^{\mathrm{op}}, \mathbf{Spc}) \\ \mathcal{C} &\mapsto [I \mapsto \mathrm{Fun}(I, \mathcal{C})^{\simeq}] \end{aligned}$$

is fully faithful [HA, Proposition A.7.10; SAG, §A.8.2; 11; 12, §1; 16]. One can explicitly identify its image; objects in the image of this embedding are often called *complete Segal spaces* or *categories internal to spaces*. Under this embedding, the subcategory $\mathbf{Spc} \subset \mathbf{Cat}_\infty$ corresponds to the constant functors $\mathbf{\Delta}^{\mathrm{op}} \to \mathbf{Spc}$. Moreover, the localization $\mathrm{B} \colon \mathbf{Cat}_\infty \to \mathbf{Spc}$ is given by geometric realization.

A.2 Notation. Let $\mathcal{C}$ be an ∞-category with finite limits. We write

$$\mathrm{Cat}(\mathcal{C}) \subset \mathrm{Fun}(\mathbf{\Delta}^{\mathrm{op}}, \mathcal{C})$$

for the full subcategory spanned by the *categories internal to* $\mathcal{C}$. See [1, Definition 13.1.1; 13, Proposition 3.2.7] for the definition.

A.3 Notation. Write $\mathbf{Cat}_{<\infty} \subset \mathbf{Cat}_\infty$ for the full subcategory spanned by those ∞-categories $\mathcal{C}$ for which there exists an integer $n \geq 0$ such that $\mathcal{C}$ is an n-category. Write $\mathbf{Cat}_{\infty,\pi} \subset \mathbf{Cat}_{<\infty}$ for the full subcategory spanned by those ∞-categories with the property that there are finitely many objects up to equivalence and all mapping spaces are π-finite.

A.4 Observation. The nerve $\mathrm{N} \colon \mathbf{Cat}_\infty \xrightarrow{\sim} \mathrm{Cat}(\mathbf{Spc})$ restricts to equivalences

$$\mathbf{Cat}_{<\infty} \xrightarrow{\sim} \mathrm{Cat}(\mathbf{Spc}_{<\infty}) \qquad \text{and} \qquad \mathbf{Cat}_{\infty,\pi} \xrightarrow{\sim} \mathrm{Cat}(\mathbf{Spc}_\pi)\,.$$

In order to describe protruncated classifying spaces via geometric realizations, it is useful to describe pro-∞-categories as category objects in prospaces.

A.5 Observation. By [HTT, Proposition 5.3.5.11; 1, Proposition 13.1.12], the composite

$$\mathbf{Cat}_{<\infty} \underset{\mathrm{N}}{\xrightarrow{\sim}} \mathrm{Cat}(\mathbf{Spc}_{<\infty}) \hookrightarrow \mathrm{Cat}(\mathrm{Pro}(\mathbf{Spc}_{<\infty}))$$

extends along cofiltered limits to a fully faithful right adjoint

$$\mathrm{N} \colon \mathrm{Pro}(\mathbf{Cat}_{<\infty}) \hookrightarrow \mathrm{Cat}(\mathrm{Pro}(\mathbf{Spc}_{<\infty}))\,.$$

This functor restricts to a fully faithful right adjoint

$$\mathrm{N} \colon \mathrm{Pro}(\mathbf{Cat}_{\infty,\pi}) \hookrightarrow \mathrm{Cat}(\mathrm{Pro}(\mathbf{Spc}_\pi))\,.$$

A.6 Remark. We do not know if the embedding $\mathrm{N} \colon \mathbf{Cat}_\infty \hookrightarrow \mathbf{Cat}(\mathrm{Pro}(\mathbf{Spc}))$ extends along cofiltered limits to a fully faithful functor $\mathrm{Pro}(\mathbf{Cat}_\infty) \to \mathrm{Cat}(\mathrm{Pro}(\mathbf{Spc}))$.

A.7 Observation. It is immediate from the definitions that the following diagram of fully faithful right adjoints commutes

$$\begin{tikzcd}
\mathrm{Pro}(\mathbf{Spc}_\pi) \arrow[r, hook, shift right] \arrow[d, hook, shift left] & \mathrm{Pro}(\mathbf{Cat}_{\infty,\pi}) \arrow[l, shift right, "\mathrm{B}^\wedge_\pi"'] \arrow[l, phantom, "\perp"] \arrow[r, hook, shift right, "\mathrm{N}"'] \arrow[d, hook, shift left] & \mathrm{Cat}(\mathrm{Pro}(\mathbf{Spc}_\pi)) \arrow[l, shift right] \arrow[l, phantom, "\perp"] \arrow[r, hook, shift right] \arrow[d, hook, shift left] & \mathrm{Fun}(\mathbf{\Delta}^{\mathrm{op}}, \mathrm{Pro}(\mathbf{Spc}_\pi)) \arrow[l, shift right] \arrow[l, phantom, "\perp"] \arrow[d, hook, shift left] \\
\mathrm{Pro}(\mathbf{Spc}_{<\infty}) \arrow[u, shift left, "(-)^\wedge_\pi"] \arrow[u, phantom, "\dashv"] \arrow[r, hook, shift right] & \mathrm{Pro}(\mathbf{Cat}_{<\infty}) \arrow[u, shift left] \arrow[u, phantom, "\dashv"] \arrow[l, shift right, "\mathrm{B}_{<\infty}"'] \arrow[l, phantom, "\perp"] \arrow[r, hook, shift right, "\mathrm{N}"'] & \mathrm{Cat}(\mathrm{Pro}(\mathbf{Spc}_{<\infty})) \arrow[u, shift left] \arrow[u, phantom, "\dashv"] \arrow[l, shift right] \arrow[l, phantom, "\perp"] \arrow[r, hook, shift right] & \mathrm{Fun}(\mathbf{\Delta}^{\mathrm{op}}, \mathrm{Pro}(\mathbf{Spc}_{<\infty}))\,. \arrow[u, shift left, "(-)^\wedge_\pi \circ -"] \arrow[u, phantom, "\dashv"] \arrow[l, shift right] \arrow[l, phantom, "\perp"]
\end{tikzcd}$$

The long composite left adjoints

$$\mathrm{Fun}(\mathbf{\Delta}^{\mathrm{op}}, \mathrm{Pro}(\mathbf{Spc}_{<\infty})) \to \mathrm{Pro}(\mathbf{Spc}_{<\infty}) \qquad \text{and} \qquad \mathrm{Fun}(\mathbf{\Delta}^{\mathrm{op}}, \mathrm{Pro}(\mathbf{Spc}_\pi)) \to \mathrm{Pro}(\mathbf{Spc}_\pi)\,.$$

are simply the colimit functors. Since the diagram of left adjoints also commutes we deduce:

A.8 Corollary. *Let $\mathcal{C} \in \mathrm{Pro}(\mathbf{Cat}_{\infty,\pi})$ be a profinite ∞-category. Then there are natural equivalences*

$$\mathrm{B}_{<\infty}(\mathcal{C}) \simeq \operatorname{colim}\left(\mathbf{\Delta}^{\mathrm{op}} \xrightarrow{\mathrm{N}(\mathcal{C})} \mathrm{Pro}(\mathbf{Spc}_{\pi}) \hookrightarrow \mathrm{Pro}(\mathbf{Spc}_{<\infty}) \right)$$

and

$$\mathrm{B}^{\wedge}_{\pi}(\mathcal{C}) \simeq \operatorname{colim}\left(\mathbf{\Delta}^{\mathrm{op}} \xrightarrow{\mathrm{N}(\mathcal{C})} \mathrm{Pro}(\mathbf{Spc}_{\pi}) \right) .$$

A.9 Example. Let $\boldsymbol{X}$ be a spectral ∞-topos. Through [1, Theorem 9.3.1], Barwick–Glasman–Haine refined the the ∞-category $\mathrm{Pt}(\boldsymbol{X})$ of points of $\boldsymbol{X}$ to a profinite ∞-category

$$\widehat{\Pi}_{(\infty,1)}(\boldsymbol{X}) \in \mathrm{Pro}(\mathbf{Cat}_{\infty,\pi})$$

called the *stratified shape* of $\boldsymbol{X}$. In [1, Theorem 10.2.3] they show that there is a natural equivalence

$$\Pi_{<\infty}(\boldsymbol{X}) \simeq \mathrm{B}_{<\infty}\left(\widehat{\Pi}_{(\infty,1)}(\boldsymbol{X})\right) .$$

That is, the protruncated shape of $\boldsymbol{X}$ can be recovered as the protruncated classifying space of the stratified shape $\widehat{\Pi}_{(\infty,1)}(\boldsymbol{X})$. Hence Corollary A.8 shows that the protruncated shape $\Pi_{<\infty}(\boldsymbol{X})$ admits a natural $\mathrm{Pro}(\mathbf{Spc}_{\pi})$-resolution in the sense of Definition 2.1.

A.10 Remark. Using the proétale topology [2], one can show that the protruncated étale homotopy type of a qcqs scheme admits a $\mathrm{Pro}(\mathbf{Set}_{\mathrm{fin}})$-resolution.

To see this, first note that given a qcqs scheme X, the pullback functor $\nu^* : X_{\text{ét}} \to X_{\text{proét}}$ from the étale ∞-topos of X to the proétale ∞-topos of X is fully faithful when restricted to truncated objects. (This observation generalizes [2, Lemma 5.1.2 & Corollary 5.1.6].) As a result, the protruncated shapes of $X_{\text{ét}}$ and $X_{\text{proét}}$ are equivalent. Hence the protruncated étale homotopy type is a hypercomplete proétale cosheaf.

Recall that the w-contractible affine schemes form a basis for the proétale topology; in particular, every qcqs scheme admits a proétale hypercover by w-contractible affine schemes. Moreover, for each w-contractible affine scheme U, the prospace $\Pi^{\text{ét}}_{<\infty}(U)$ is naturally identified with the profinite set $\pi_0(U)$ of connected components of U. Hence the protruncated étale homotopy type

$$\Pi^{\text{ét}}_{<\infty} : \mathbf{Sch}^{\mathrm{qcqs}} \to \mathrm{Pro}(\mathbf{Spc}_{<\infty})$$

is the unique hypercomplete proétale cosheaf whose restriction to w-contractible affines is given by $U \mapsto \pi_0(U)$. Given a qcqs scheme X, choose a proétale hypercover $U_\bullet$ of X be w-contractible affines. Then the simplicial object $\pi_0(U_\bullet)$ is a $\mathrm{Pro}(\mathbf{Set}_{\mathrm{fin}})$-resolution of $\Pi^{\text{ét}}_{<\infty}(X)$.

References

HTT J. Lurie, *Higher topos theory*, Annals of Mathematics Studies. Princeton, NJ: Princeton University Press, 2009, vol. 170, pp. xviii+925, ISBN: 978-0-691-14049-0; 0-691-14049-9.

HA ______, *Higher algebra*, Sep. 2017, `math.ias.edu/~lurie/papers/HA.pdf`.

SAG ______, *Spectral algebraic geometry*, Feb. 2018, `math.ias.edu/~lurie/papers/SAG-rootfile.pdf`.

Ker ______, *Kerodon*, `kerodon.net`, 2024.

1. C. Barwick, S. Glasman, and P. J. Haine, *Exodromy*, Aug. 2020, `arXiv:1807.03281v7`.

2. B. Bhatt and P. Scholze, *The pro-étale topology for schemes*, Astérisque, no. 369, pp. 99–201, 2015, arXiv:1309.1198.

3. T. Blom and I. Moerdijk, *Profinite* ∞*-operads*, Adv. Math., vol. 408, no. part B, Paper No. 108601, 50, 2022. DOI: 10.1016/j.aim.2022.108601, arXiv:2107.10092.

4. ______, *Profinite* ∞*-operads: Part II*, Dec. 2023, arXiv:2312.12567.

5. P. Boavida de Brito, G. Horel, and M. Robertson, *Operads of genus zero curves and the Grothendieck–Teichmüller group*, Geom. Topol., vol. 23, no. 1, pp. 299–346, 2019. DOI: 10.2140/gt.2019.23.299, arXiv:1703.05143.

6. B. Calmès, E. Dotto, Y. Harpaz, F. Hebestreit, M. Land, K. Moi, D. Nardin, T. Nikolaus, and W. Steimle, *Hermitian* K*-theory for stable* ∞*-categories I: Foundations*, Selecta Math. (N.S.), vol. 29, no. 1, Paper No. 10, 269, 2023. DOI: 10.1007/s00029-022-00758-2, arXiv:2009.07223.

7. C.-Y. Chough, *Proper base change for étale sheaves of spaces*, J. Eur. Math. Soc. (JEMS), vol. 24, no. 11, pp. 3679–3702, 2022. DOI: 10.4171/jems/1263.

8. P. J. Haine, T. Holzschuh, and S. Wolf, *Nonabelian basechange theorems & étale homotopy theory*, Apr. 2023, arXiv:2304.00938.

9. ______, *The fundamental fiber sequence in étale homotopy theory*, Int. Math. Res. Not. IMRN, no. 1, pp. 175–196, 2024. DOI: 10.1093/imrn/rnad018, arXiv:2209.03476.

10. L. Hesselholt and P. Pstrągowski, *Dirac geometry I: Commutative algebra*, Peking Math. J., 2023. DOI: 10.1007/s42543-023-00072-6, arXiv:2207.09256.

11. A. Joyal and M. Tierney, *Quasi-categories vs Segal spaces*, in *Categories in algebra, geometry and mathematical physics*, Contemp. Math. Vol. 431, Amer. Math. Soc., Providence, RI, 2007, pp. 277–326. DOI: 10.1090/conm/431/08278, arXiv:0607820.

12. J. Lurie, $(\infty, 2)$*-Categories and the Goodwillie Calculus I*, Oct. 2009, math.ias.edu/~lurie/papers/GoodwillieI.pdf.

13. L. Martini, *Yoneda's lemma for internal higher categories*, Apr. 2022, arXiv:2103.17141.

14. M. C. McCord, *Singular homology groups and homotopy groups of finite topological spaces*, Duke Math. J., vol. 33, pp. 465–474, 1966.

15. T. Oda, *Étale homotopy type of the moduli spaces of algebraic curves*, in *Geometric Galois actions, 1*, London Math. Soc. Lecture Note Ser. Vol. 242, Cambridge Univ. Press, Cambridge, 1997, pp. 85–95.

16. C. Rezk, *A model for the homotopy theory of homotopy theory*, Trans. Amer. Math. Soc., vol. 353, no. 3, pp. 973–1007, 2001. DOI: 10.1090/S0002-9947-00-02653-2, arXiv:9811037.

17. M. Volpe, *Verdier duality on conically smooth stratified spaces*, Jun. 2022, arXiv:2206.02728.